Final exam of Biochemistry

Fill-in the blank questions

1) In a protein chain three dihedral angles are defined as___ angles. (answer: ϕ, phi; ψ, psi; and ω, omega)
2) An___ion is a hydrogen ion associated with one or more molecules of water. (answer: hydronium)
3) Disulfide bonds between cysteinyl residues can be__ by mercaptans. (answer: reduced)
4) Supersecondary structures, also called___, occur in many unrelated globular proteins. (answer: motif or fold)
5) The oxygen binding curve of hemoglobin is ____. (answer: sigmoidal)
6) A sugar forms a cyclic-hemiacetal by reaction between an alcohol and an ____. (answer: aldehyde)
7) ___ base pairs can occur at the third position of the codon, as predicted by the 'wobble' hypothesis. (answer: Non Watson-Crick)
8) ATP is considered an "energy rich" compound because it contains two _____ (answer: phosphoanhydride) bonds.
9) At the ribosome an mRNA codon must pair with the anticodon of the correct ____. (answer: aminoacyl-tRNA)
10) Phototrophs are organisms that use energy from____ (answer: sunlight) and convert it to chemical energy.

Multiple choice questions

1. It is believed that life evolved with RNA as the genetic material, but RNA has been replaced by DNA in all current cellular life. Which feature of DNA probably accounts for this? (answer is C)

a) DNA can direct its own replication while RNA cannot.

b) DNA is a nucleic acid while RNA is not.

c) DNA is more stable than RNA.

d) Only DNA can form the genetic material of viruses.

2. A ribozyme is: (answer is A)

a) a class of RNA molecule that can catalyse chemical reactions.

b) a monomeric subunit of RNA.

c) a particle composed of RNA and protein that is involved in the synthesis of proteins.

d) a protein enzyme that catalyses the synthesis of RNA.

3) ATP (adenosine triphosphate) can be termed the universal energy currency in life because it: (answer is d)

a) enables cells to disobey the first law of thermodynamics.

b) enables cells to disobey the second law of thermodynamics.

c) releases energy as heat when it undergoes hydrolysis.

d) traps energy released from the oxidation of different classes of food molecules.

4) Which term below best defines the 'quaternary structure' of a protein? (answer is a)

a) The arrangement of two or more polypeptide subunits into a single functional complex

b) The folding of the polypeptide backbone in three-dimensional space.

c) The interaction of amino acid side chains.

d) The sequence of amino acids in a polypeptide chain.

5） Which of the following most accurately describes how secondary structures in proteins are stabilized? (Answer is d)

a) Through ionic bonds operating between oppositely charged amino acid side chains.

b) Through covalent bonds joining different parts of the peptide backbone.

c) Through hydrogen bonds between different amino acid side chains.

d) Through hydrogen bonds joining different parts of the peptide backbone.

6) Which of the following amino acids is mostly likely to disrupt an alpha helix? (answer is a)

a) Proline.

b) Leucine.

c) Glycine.

d) Valine.

7) How many different transfer RNA molecules are present in a cell (not including those present in the mitochondria)? (answer is d)

a) 64

b) 61

c) 20

d) More than 20, less than 61.

8) Proteasomes destroy proteins by unfolding them and chopping them into small fragments. What kinds of proteins are recognized by proteasomes? answer is a

a) Proteasomes recognise ubiquitinylated proteins.

b) Proteasomes recognise denatured proteins.

c) Proteasomes recognise proteins with basic or aromatic amino acids at the N-terminus.

d) Proteasomes recognise viral proteins.

9) Which of the following statements about the use of mass spectrometry in protein investigation are correct? Please select all that apply. Answers are all.

a) Mass spectrometry is used to rapidly determine the molecular weight of a protein.

b) Mass spectrometry can be used to determine the sequence of peptides or proteins.

c) Mass spectrometry can be used to investigate post-translational modifications.

d) Mass spectrometry may involves separating ionic fragments on a gel.

10) Which of the following statements about protein separation by gel filtration are correct? Please select all that apply. Answers are b and c

a) When a mixture of proteins is separated by gel filtration, the smallest molecular weight protein is eluted first.

b) When a mixture of proteins is separated by gel filtration, the largest molecular weight protein is eluted first.

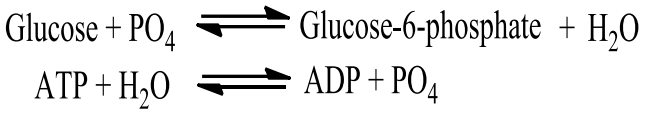
c) When a mixture of proteins is separated by gel filtration, the smallest molecular weight protein is eluted last.

d) When a mixture of proteins is separated by gel filtration, the smallest molecular weight proteins flow around the beads.

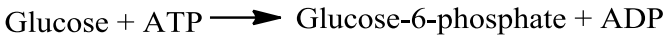
Cumulative

1) Glucose-6-phosphate is formed from glucose in an ATP-utilizing reaction.

a) Using the following information, calculate the overall $\Delta G^{0'}$ for this reaction. (answer: $\Delta G^{0'}$= -20.9 kJ/mol)



b) Write out the overall reaction

Aswer: 

c) In which direction will the reaction proceed? Answer: right

d) Given your answer in a), is the overall reaction **spontaneous** or **not spontaneous**? Answer: Spontaneous

2) Soy sauce is one common ingredient in Philippine cooking. It is made by fermenting a mixture of mashed soybeans (mostly protein) and flour (mostly starch), followed by brining (soaking the fermented mixture in salt water). The brining liquid becomes the soy sauce. (10 points, 2 points/each)

   a) To what class of macromolecules does starch belong? Answer: Carbohydrates

   b) What are the structural features of the polymers in starch? Choose all that apply: b1) Linear; b2) Branched; b3) Helical; b4) Extended; b5) Rigid; b6) Hydrated; b7) Anhydrous; b8) Fibrous

   Answer: b1, b2, b3, b6

   c) What are the monomer units of starch? Be as specific as possible Answer: α-D-glucopyranose

   d) Name the linkages that connect these monomers in starch. Be as specific as possible

   Answer: α-1,4 and α-1,6 O-glycosidic linkage

   e) Briefly state the most likely purpose of adding flour to the soybeans (10 words or fewer.)

   Answer: To provide a sugar source for fermentation